

# CS 307: Software Engineering

## Homework #2

Due on March 5th, 2026 at 11:58pm  
CS 307: Software Engineering Homework #2

### Problem 1

This is a group homework problem. Your group will be same as the one that you are enrolled in for the team project in this course.

Find one open-source project that is closest to your team project. Reverse engineer the design and a few architectural/design patterns from that project.

Describe the above reverse engineered details in the following format. If applicable, provide pointers to project's directories/files/classes in your descriptions and/or diagrams.

#### 1. Purpose (1 point)

- a. Briefly explain the chosen open-source project and its purpose.
- b. Provide the link to the open-source project repository.

#### 2. Design Outline (2 points)

- a. Identify its components and describe the purpose of each component.
- b. Describe the interactions between components.
- c. Include at least one UML diagram that clearly shows high-level structure of the system.

#### 3. Design Details (7 points)

- a. Include class level design (i.e., class diagrams) and be as detailed as you can.
- b. Describe the classes and interactions between the classes.
- c. Include sequence diagrams for different activities and/or activity diagrams and/or state diagrams (at least five UML diagrams).
- d. In your descriptions and/or diagrams, point to the source file and line numbers if applicable, so that the grader can have a look.

#### 4. Design Patterns (4 points)

- a. Provide textual description of the patterns along with their diagrams (at least five patterns).
- b. In your descriptions and/or diagrams, point to the source file and line numbers if applicable, so that the grader can have a look.

#### 5. Design Issues (1 point)

- a. Identify one functional design issue and one non-functional design issue.
- b. Each design issue requires a descriptive title, the solution option chosen by the

project, and the alternative solution that you can think of which will either degrade/improve the design.

## **Submission**

Please make sure to include team number and the names of all your team members. You are expected to follow the organization of the design documentation with a right information in the right section and appropriate pointers to the project repository content in your descriptions and/or diagrams.

A homework assignment will be created on Gradescope (only one submission per group). You are expected to submit a document in PDF format.

Violation of the above instructions will result in a reduction in score.

